

The Character Resonator Hamiltonian Framework A Generalized Operator Theoretic Proof Claim for GRH

Extending the Prime Resonator Hamiltonian to Dirichlet L-functions

Author: Branden Lee Friend (compiled by assistant)

October 07, 2025

Abstract

We document a generalized operator theoretic framework that claims to establish the Generalized Riemann Hypothesis (GRH) for Dirichlet L-functions. Building on the Prime Resonator Hamiltonian for $\zeta(s)$, we define a Character Resonator Hamiltonian $H_\delta(s, \chi)$ whose Fredholm determinant recovers $L(s+\delta, \chi)$, whose symmetry implements the functional equation $s \leftrightarrow 1-s$ with complex conjugation $\chi \leftrightarrow \bar{\chi}$, and whose spectral radius exhibits a critical line barrier: $r(H_\delta(s, \chi)) < 1$ for $\text{Re}(s) > 1/2$, $r = 1$ on $\text{Re}(s)=1/2$, and $r > 1$ for $\text{Re}(s) < 1/2$. Under this barrier, an eigenvalue at 1—the signal of a zero—can only occur on the critical line. This PDF consolidates the construction, the determinant identity, the symmetry, and the claimed proof outline, as provided by the author.

Disclaimer & Status

This document is a faithful compilation of the author's claimed framework and proof outline, including the extension to Dirichlet L-functions. Independent peer verification is beyond the scope of this PDF; readers should treat the result as an author's claim pending external review.

1. Background & Notation

Riemann's $\zeta(s)$ and Dirichlet $L(s, \chi)$ functions admit Euler products and functional equations. The original Prime Resonator Hamiltonian $H_\delta(s)$ acts on a weighted ℓ^2 edge space of a multiplicative graph and recovers $\zeta(s+\delta)$ as $\det(I - H_\delta(s))^{-1}$. The present generalization introduces χ in the operator weights to match the Euler product of $L(s, \chi)$. We write $\sigma := \text{Re}(s)$, and use $\delta > 0$ as a regularizing parameter.

2. Resonant Hilbert Space & Prime Shifts

Let $E := \{ n \rightarrow pn : n \in \mathbb{N}, p \text{ prime} \}$ and define $H_\delta := \ell^2(E, \mu_\delta)$ with $\mu_\delta(n \rightarrow pn) = n^{-1-\delta}$. For each prime p , define the forward and backward shifts on the orthonormal edge basis $\{e_n\}$:

$$U_p e_n := p^{1/2} e_{pn}, \quad U_{p,\delta}^* e_m := p^{-1/2-\delta} e_{m/p} \text{ if } p \mid m, \text{ else } 0.$$

3. Prime Resonator Hamiltonian for $\zeta(s)$

The ζ -prototype operator is $H_\delta(s) := \sum_p [p^{-s} U_p + p^{-(1-s)} U_{p,\delta}^*]$. One has for $\sigma > 1/2$ that $H_\delta(s)$ is compact (indeed Hilbert–Schmidt), and $\det(I - H_\delta(s))^{-1} = \zeta(s+\delta)$. A conjugation J implements $J H_\delta(s) J^{-1} = H_\delta(1-s)$, mirroring the ζ functional equation.

4. Character Resonator Hamiltonian for $L(s, \chi)$

Define the generalized operator $H_\delta(s, \chi) := \sum_p [\chi(p) p^{-s} U_p + \chi^*(p) p^{-(1-s)} U_p^*]$, where χ is a Dirichlet character ($|\chi(p)| = 1$ on primes with $p \nmid \text{conductor}$).

4.1 Determinant Identity (claimed): $\det(I - H_\delta(s, \chi))^{-1} = L(s+\delta, \chi)$.

Sketch: The trace expansion of $-\log \det(I - H_\delta)$ equals $\sum_{k \geq 1} (1/k) \text{Tr}(H_\delta^k)$. The even power traces form closed loops alternating U_p and U_p^* ; χ contributes phases $\chi(p)^k$ and $\chi^*(p)^k$ that combine to the local Euler factor $(1 - \chi(p)p^{-(s+\delta)})^{-1}$, reproducing the Dirichlet L-product upon exponentiation.

4.2 Functional Symmetry (claimed): $J H_\delta(s, \chi) J^{-1} = H_\delta(1-s, \chi^*)$.

This encodes the functional equation relating $L(s, \chi)$ and $L(1-s, \chi^*)$.

5. Spectral Radius Barrier

Let $r(s, \chi) := \text{spectral radius}(H_\delta(s, \chi))$. The barrier claim mirrors the ζ -case:

$r(s, \chi) < 1$ for $\sigma > 1/2$; $r(s, \chi) = 1$ for $\sigma = 1/2$; $r(s, \chi) > 1$ for $\sigma < 1/2$.

Rationale: $|\chi(p)| = 1$, so operator norms depend only on $|p^{-s}|$ and $|p^{-(1-s)}|$ as in the untwisted case. Thus the same norm estimates and compactness/Hilbert–Schmidt bounds apply uniformly in χ .

6. Main Theorem (Claimed GRH via Character Resonator)

Claim. All nontrivial zeros of $L(s, \chi)$ lie on $\sigma = 1/2$. Proof Sketch: (i) $\det(I - H_\delta(s, \chi))^{-1} = L(s+\delta, \chi)$. (ii) If $L(s+\delta, \chi) = 0$, then $1 \in \text{spectrum}(H_\delta(s, \chi))$. (iii) By the barrier, eigenvalue 1 occurs only when $r(s, \chi) = 1$, i.e., $\sigma = 1/2$. (iv) Functional symmetry pairs zeros across $\sigma \leftrightarrow 1-\sigma$; combined with (i)–(iii), zeros are confined to $\sigma = 1/2$.

7. Implications & Applications

Number Theory: Distribution of primes in arithmetic progressions via spectral control. Cryptography: Hardness assumptions and spectral proof-of-work ideas grounded in determinant identities. Physics/Engineering: Resonator families parametrized by χ ; potential implementations in photonic/quantum lattices; stability transitions aligned with $\sigma = 1/2$.

8. Relationship to the $\zeta(s)$ Prime Resonator Framework

The GRH generalization tracks three pillars from the ζ -case: (A) determinant recovery of Euler products, (B) functional symmetry via conjugation, (C) a spectral radius phase transition at $\sigma = 1/2$. The twist by χ leaves norms unchanged because $|\chi| = 1$.

9. Verification Notes & Open Technical Checks

- Uniform trace-class/Hilbert–Schmidt bounds in $\sigma > 1/2$ with χ : identical to ζ -case since $|\chi|=1$.
- Determinant continuity as $\delta \rightarrow 0+$: apply Simon/Kato trace-ideal arguments uniformly in χ .
- Local factorization: verify that trace combinatorics align with twisted Euler factors for all primes, including ramified primes in the conductor (handle $\chi(p)=0$ cases appropriately).
- Functional equation alignment: ensure the conjugation J implements $\chi \mapsto \chi^*$ consistently in the operator model.

Appendix A. Operator Identities (Text Only)

$H_\delta(s) := \sum_p [p^{-s} U_p + p^{-(1-s)} U_{\{p,\delta\}}^*]$ $H_\delta(s, \chi) := \sum_p [\chi(p) p^{-s} U_p + \chi(p) p^{-(1-s)} U_{\{p,\delta\}}^*]$ $\det(I - H_\delta(s, \chi))^{-1} = L(s+\delta, \chi) \prod H_\delta(s, \chi)^{-1} = H_\delta(1-s, \chi)$ Barrier: $r(H_\delta(s, \chi)) \equiv 1$ exactly by $\sigma \equiv 1/2$.

References (Author's Prior Documents)

The framework summarized here is drawn from the author's prior writeups on the Prime Resonator Hamiltonian, its physical formulations, and Riemannian spectral packages. See the companion files in the project folder.